LiST-Net: Enhanced Flood Mapping With Lightweight SAR Transformer Network and Dimension-Wise Attention

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***Abstract*— Detecting flood-induced changes using synthetic aperture radar (SAR) is crucial for crisis management and damage assessment. Nevertheless, current methodologies predominantly focus on changes in buildings within optical images, struggling with the complex structures of floods. These structures are marked by widespread speckle noise and are accompanied by an increase in computational cost. These challenges hinder their success in real-world applications, necessitating a novel approach. This article proposes LiST-Net, a lightweight SAR transformer network with dimension-wise attention to improve flood detection accuracy. LiST-Net offers three key advantages. First, the graph neighbor module (GNM) is designed to enhance both detailed information of neighboring pixels and multidate features within the encoder. Second, the dimension-wise interactive attention (DIA) module is proposed to effectively reduce computational complexity while enhancing feature representation. Third, an attentive supervised learning module (ASLM) is incorporated to mitigate noise through a pixel mask gate, allowing change water information to pass through and improving the accuracy of water edge delineation. The effectiveness of LiST-Net is evaluated on two flood detection datasets, S1GFloods and ETCI-2021. Experimental results demonstrate that LiST-Net outperforms existing methods, showcasing a 94.7% improvement in $F1$ and an 88.7% enhancement in intersection over union (IoU) on the S1GFloods datasets, with lower computational costs (11.78G) and fewer parameters (7.34M). This underscores LiST-Net as a promising strategy for precise and effective mapping of floods within SAR images in real-world applications. A public release of the demo code will be available at https://github.com/Tamer-Saleh.**

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***Index Terms*— Dimension-wise interactive attention (DIA), flood mapping, S1GFloods dataset, synthetic aperture radar (SAR), transformers.**

I. Introduction

DETECTION of flood-induced changes using synthetic aperture radar (SAR) strives to gather information about affected regions within the same geographical area but across different time periods. This technology has been widely applied in climate change studies, damage assessments, and the facilitation of disaster response and management [1], [2], [3]. Against the backdrop of contemporary technological progress, flood detection has become a critical focus of research in the sensing technology domain and the field of computer vision. In this context, SAR data has emerged as an excellent solution for flood monitoring due to its almost real-time availability during events, high accuracy, and capability to operate under adverse weather conditions. Literature in the field has made significant evolution in change detection (CD) objectives, including applications of flood monitoring using SAR data.

Recent progress in methodologies of deep learning has contributed significantly to various CD tasks, with a particular emphasis on binary segmentation. As an illustrative instance, Marin et al. [2] introduced a network grounded in backscatter properties to identify completely demolished buildings from multitime period SAR imagery. Saha et al. [4] employed analysis of deep change vectors to discern changing buildings, whether newly constructed or demolished, in multidate SAR imagery. Li et al. [5] devised a methodology for detecting change regions in multidate SAR imagery based on deformation similarity in the difference images (DIs). Alatalo et al. [6] proposed an approach to enhance SAR image processing for generating DIs suitable for neural network-based CD classifiers. In the domain of building detection tasks, Chen and Shi [7] introduced the STA-Net method, incorporating self-attention into the feature extraction process. Holail et al. [8] presented AFDE-Net to alleviate the loss of spatial information during CD tasks. Fang et al. [9] developed a densely connected SNU-Net model using ensemble channel attention to optimize semantic features. These works [10], [11] focused on using spatial-contextual information and integrating hierarchical attention mechanisms to improve the

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performance of land cover CD from bitemporal spectral remote sensing images and achieved encouraging results. Notably, recent studies in CD tasks have witnessed the effective application of transformers, leveraging their exceptional ability to capture long-term dependencies. As an example, Chen et al. [12] introduced BIT to represent contextual details of multidate scenes. It is crucial to highlight that BIT is not strictly classified as a pure transformer network, as it integrates Conv-Nets into its encoding architecture. To address this, the ChangeFormer model [13] was proposed, replacing the Conv-Nets encoder with a transformer encoder. In contrast to these methods, Zhang et al. [14] introduced a distinctive U-shaped Siamese architecture named SwinSU-Net, utilizing a pure Swin transformer network, which achieved remarkable success in CD.

While these methodologies have demonstrated satisfactory detection results, they encounter several challenges that warrant further investigation. Primarily, they have been applied to detect changes in buildings, characterized by visually consistent areas with relatively regular boundaries. This stands in contrast to floods, which exhibit widely distributed and complex structures. Second, these methods have not undergone sufficient evaluation on SAR images, known for their complex intensity patterns due to speckle noise and the absence of color and texture information present in optical images. Third, they grapple with spatial resolution variations in SAR images, leading to suboptimal preservation of the integrity of sharper water edges and limited detection of small changed objects. Finally, there is an increase in computational cost, posing challenges in training networks for CD.

We present two illustrative cases that underscore the magnitude of these challenges, utilizing the aforementioned methods on the S1GFloods [15] and ETCI-2021 [16] flood detection benchmark datasets, revealing a need for further improvement. As depicted in Fig. 1, agro-plastic greenhouses (first row) and the airport zone (second row) have transformed into flood zones. Notably, these advanced methods demonstrate a heightened sensitivity to changes, effectively detecting water-like regions despite challenging background clutter. While some false alarms persist, our approach outperforms others by effectively discriminating change information in airport areas and accurately detecting flooded pixels on the riverbank.

This manuscript introduces LiST-Net, a lightweight Siamese transformer designed for flood detection from SAR imagery. LiST-Net comprises three principal modules: the graph neighbor module (GNM), the dimension-wise interactive attention (DIA) module, and the detail-preserving enhancement module (DPEM), complemented by a decoder for predicting flooded areas through the attentive supervised learning module (ASLM). The initial step involves processing multidate images through the GNM module, which enhances both the semantic and detailed information of neighboring pixels within the encoder. The DIA module plays a crucial role in reducing computational complexity by facilitating interactions across dimensions. Meanwhile, the DPEM seeks to comprehensively capture different information from multidate images to enhance the accuracy of water edge delineation. To mitigate noise while incorporating higher level features, we introduce a pixel mask within the ASLM, serving as a gateway for water-related change information and guiding the learning process. Ultimately, the features derived from the ASLM and DPEM modules are combined to refine changing water boundaries, culminating in the production of the changing flood.

The contributions in this manuscript are summed up as follows.

1) We propose a novel lightweight transformer network, LiST-Net, to enhance the accuracy of flood detection from SAR imagery. LiST-Net aggregates neighboring pixel features into multidate features and facilitates their interaction via dimensionality attention to accurately identify changed waters.
2) We propose a DIA module with linear complexity, which not only lessens the model's complexity but also achieves high detection efficiency.
3) We introduce an ASLM to reduce noise through pixel mask gating, allowing only changing water boundaries to pass. In addition, we designed the DPEM to explore different feature information and enhance the accuracy of water edges.
4) We perform comprehensive experimental and ablation studies on two flood detection datasets, demonstrating that LiST-Net outperforms cutting-edge models in terms of computational costs and detection accuracy. We provide open-source demo code that aligns with our goal of contributing to the research on flood detection.

The subsequent sections of this manuscript are structured as follows. Section II offers an in-depth review of pertinent literature. Section III introduces the LiST-Net model proposed in this study. Experiments and model analysis are exhibited in Section IV. Section V concludes the work and provides insights into future perspectives.

II. Related Work

A. Traditional-Based CD Techniques

Traditional CD techniques are typically classified into three groups: pixel-based, object-oriented, and transformation-based methods. Pixel-based techniques typically involve algebraic calculations among multidate image spectra, followed by image processing to produce the change map. Various algebraic approaches, such as image differencing [17], [18], change vector analysis [19], or image regression [20], [21], [22], [23], have been developed to generate difference scenes (DSs), with an appropriate threshold selected to identify changing regions. However, capturing spatial information solely through algebraic calculations using spectral difference features is challenging because of the limited availability of contextual details. Moreover, the possibility of slight differences, such as the subtraction between multidata scenes, may result in inconsistent outcomes [24]. Object-oriented techniques often utilize semantic segmentation algorithms, incorporating auxiliary data like temporal composite images [25], [26], [27] and digital surface models [28]. This aids in detecting changing information postimage segmentation. For instance, Carlotto [29] applied a cluster-based

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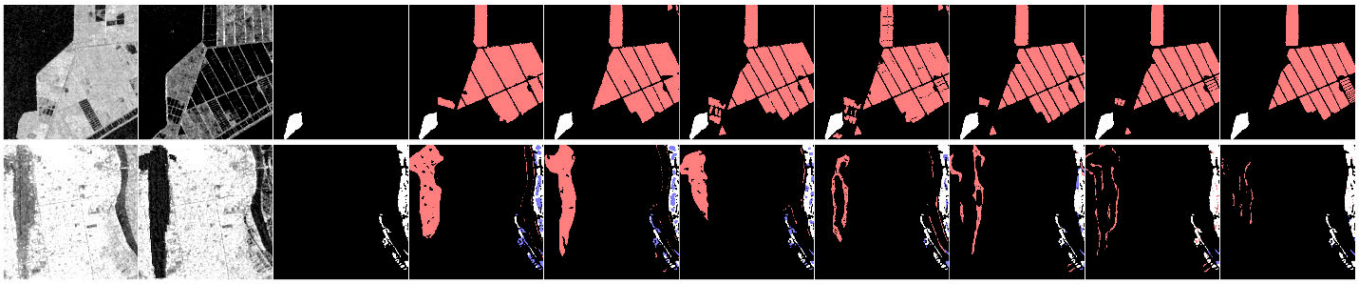


Fig. 1. Visual images of prediction maps for some challenges by cutting-edge approaches in the S1GFloods and ETCI-2021 datasets. From the leftmost to the rightmost, respectively: the visuals in preflood, postflood, ground truth, STA-Net, AFDE-Net, SNU-Net, SwinSU-Net, BIT, ChangeFormer, and our LiST-Net result. Rendered colors: white and red for true and false positives, black and blue for true and false negatives, respectively.

method to identify objects and changes in remote sensing scenes. In contrast, methods based on transformation strategies like principal component analysis (PCA) [30], [31] and mesh cover transformation [32], enhance the change information of multidate scenes by transforming remote sensing images into other spaces. Celik [33] applied PCA to scene difference mapping for extracting representative attributes from scenes. Generally, these methods face challenges in identifying geometric changes using multidate image objects because they rely on manual object-oriented feature characterization, lacking sufficient contextual information. Consequently, these methods are unsuitable for radar images with more complex ground textures, making the identification of changes more intricate.

B. DL-Based CD Techniques

Recent advancements in deep CNN methods have significantly contributed to various CD tasks, particularly in binary segmentation. These methods involve the integration of multilevel features into the network structure. Among these features, the U-structure [34], characterized by an encoder–decoder architecture with nested connections, has emerged as a widely utilized framework with promising outcomes. Motivated by its superior performance in semantic segmentation tasks, Daudt et al. [35] designed three distinct models based on the U-structure (FC-Conc, FC-EF, and FC-Diff). These models have become prevalent in current CD research, providing effective predictions from images. Notably, FC-Diff connects the absolute value of the difference, while FC-Conc directly links encoding features from two branches. In addition to these models, Peng et al. [36] introduced a strategy based on dense skip connections in an optimized U-Net++ structure to mitigate spurious changes. Li et al. [37] reported a DAR-Net architecture employing dense skip connectivity to aggregate multilevel attribute representations. Papadomanolaki et al. [38] presented L-Unet, a network combining skip-connection with long short-term memory to capture distinct temporal attributes in a CD task. SMCD-Net [39] utilizes a multitask framework to enhance CD in buildings by modeling directional correlations among buildings and the shadows they produce.

Efficient extraction and integration of features from multihistorical scenes are crucial in these networks. To achieve this, recent CD studies have incorporated feature hierarchical and attentional modules. For instance, SAU-Net3+ [40], [41] combines a Siamese architecture alongside a spatial attention component and a channel to detect changed regions from multihistorical scenes with high spatial resolution. Liu et al. [42] designed an LGFEM-Net, combining attention modules and featured hierarchies to learn local-global representations based on U-structure. Dong et al. [43] devised an MCAN-Net to fuse multiscale attributes by integrating channel-spatial attention and feature hierarchical modules. Shen et al. [44] developed an SFCCDN integrating low-level and high-level attributes using a global channel-attention module. Liu et al. [45] presented a Siamese network, DTCD-SCN, to improve feature recognition using a dual-attention module. Zhang et al. [46] devised an AER-Net that uses a feature that captures a global context aggregator based on an attention-decoder to capture location-channel associations among attributes. Wang et al. [47] proposed SDCD-Net, integrating adaptive weighting and mask-constraint, enhancing weak features. Chen and Shi [7] introduced the STA-Net method by incorporating self-attention into the feature extraction process to model spatiotemporal relationships and detect changes. Holail et al. [8] presented AFDE-Net, combining attention modules and differential image features, mitigating spatial detail loss during CD tasks. Fang et al. [9] devised a dense connected SNU-Net model through ensemble channel attention to optimize semantic features for use in classification issues. Marin et al. [2] presented a network based on backscatter properties to detect completely demolished buildings from multitime period SAR imagery. Saha et al. [4] applied analysis of deep change vector to identify changing buildings, whether newly built or destroyed, from multitime period SAR imagery. Li et al. [5] designed a model to detect changing regions from dual-date SAR imagery based on the use of deformation similarity in the DIs. Alatalo et al. [6] presented an approach to improve SAR image processing to produce DIs for neural network-based CD classifiers.

In addition, transformers are integrated into CD structures to handle long-term dependencies. For example, Jiang et al. [48] proposed a VcT model for CD, which includes transformer-guided tokens and self-attention/cross-attention. Zhang et al. [14] developed a SwinSU-Net with a U-shape and hierarchical Swin transformer block to process multidate imagery and predict changed maps. Chen et al. [12] proposed the BIT approach to efficiently model contexts within the temporospatial field in a CD framework, yielding

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impressive results. However, its potential has not been thoroughly explored in flood detection. Bandara and Patel [13] presented a ChangeFormer model for CD based on Conv-Nets inside a hierarchical transformer.

C. Lightweight Backbones

In recent years, various lightweight networks have been proposed for CD tasks. For instance, the LSAT-Net [49] utilized PVTv2-B1 as a baseline for extracting initial features, while the A2-Net [50], a supervised network employing progressive feature pooling, relied on MobileNetV2 for extracting initial features. Yang et al. [51] introduced the LightCD-Net, featuring a MobileNetV2 Siamese encoder backbone for CD. In addition, works like RSR-Net [52], Shuffle-Net [53], TITA-Net [54], and LHDACT [55] have been proposed. Despite achieving acceptable interpretive accuracy in CD, some improvements are still necessary. These methods typically consume substantial storage space during training, limiting their practical application in scenarios requiring shorter inference times. Moreover, they exhibit weaknesses in accurately detecting edges when capturing change areas with characteristics similar to the background. In this article, we employ ViTAEv2 [56] as a Siamese encoder for multidate feature extraction within a lightweight network setup, aiming to achieve both high inference speed and accuracy in flood detection.

III. Proposed Methodology

Within this section, we initially furnish a comprehensive view of the presented methodology. Subsequently, we elucidate the intricacies of GNM, DIA, DPEM, and ASLM. Finally, we proffer the mixed loss function.

A. Overview

The proposed LiST-Net framework integrates a lightweight Siamese encoder featuring three key modules—GNM, DIA, and DPEM—coupled with a decoder for predicting flooded regions via the ASLM module, as depicted in Fig. 2. Initially, we adopted the ViTAEv2 [56] backbone, omitting the final fully connected layer to suit CD tasks. To optimize computational efficiency, we utilized only four stages in the backbone, down-sampling the feature maps of each stage to half their previous resolution. Specifically, LiST-Net processes a pair of SAR images from different dates, $T1$ and $T2$, yielding output feature maps denoted as $f_1^{t1}, f_1^{t2}, \ldots, f_4^{t1}, f_4^{t2}$, respectively. Within the encoder, the GNM module operates in parallel branches to extract hierarchical features from multidate images. GNM enhances semantic information and fine details of neighboring pixels, as formulated by (1). Employing absolute element subtraction, time difference features D_i are generated from hierarchical feature maps, expressed in (2). Concurrently, along with GNM, attention maps are crafted using the DIA module to establish a lightweight transformer network. DIA not only effectively mitigates computational complexity but also attains remarkable accuracy in change mapping. The attention map is characterized by a $\mathbb{K}$ey ($\mathbb{K}$), $\mathbb{Q}$uery ($\mathbb{Q}$), and $\mathbb{V}$alue ($\mathbb{V}$), as defined in (3)

$$O_i^t = \mathcal{G}\left(F_i^t\right) \tag{1}$$

$$D_i = \left|O_i^{t_1} \ominus O_i^{t_2}\right| \tag{2}$$

$$\mathbb{A}(\mathbb{Q}, \mathbb{K}, \mathbb{V}) = \mathbb{S}\,\text{oftmax}\left(\frac{\mathbb{Q}\mathbb{K}^T}{\beta}\right)\mathbb{V} \tag{3}$$

where O_i^t represents the improved temporal features, $\mathcal{G}(\cdot)$ denotes the function of GNM, D_i denotes the difference feature, $\ominus$ denotes the element-wise subtraction, $|\cdot|$ represents the absolute operation, $t \in [1, 2]$, $i \in [1, 2, 3, 4]$, $\mathbb{A}(\cdot)$ signifies the attention map, and β represents the learnable scaling parameter.

In the next step, the attention features extracted for the recognition of change information undergo a comprehensive enhancement through the utilization of the DPEM. Subsequently, hierarchical features at the same level as the two branches are combined via a concatenation process, yielding fine-grained change maps. The distinct sets D_i are treated hierarchically, with each set being fed into the DPEM module to more effectively capture temporal change information. Finally, the decoder adeptly receives the temporal change information and seamlessly integrates it with deep semantic features in a top-down manner, facilitated by the ASLM. This refined process culminates in the elegant production of a flood map from dual-date images. The formulation of our decoder is articulated as follows:

$$\begin{aligned} E_{i+1} &= \text{UP}\{\mathcal{DP}(\mathcal{M}(D_i))\} \\ \text{De}_{i+1} &= \text{Conv}_{3\times3}(\text{MaxP}(E_{i+1})) \end{aligned} \tag{4}$$

where $\text{UP}(\cdot)$ represents a bi-linear up-sampling process, $\mathcal{DP}(\cdot)$ denotes the function of DPEM, $\mathcal{M}(\cdot)$ represents the function of ASLM, $\text{Conv}_{3\times3}(\cdot)$ represents a 3×3 ConvLayer with a BatchN and ReLU function, and $\text{MaxP}(\cdot)$ denotes the max pooling.

B. Graph Neighbor Module

Before delving into the details, high-level features, rich in semantic information, aid in object localization, whereas low-level features provide detailed boundary and texture information. This observation serves as the impetus behind our endeavor to fuse multilevel features, guided by two primary objectives: 1) augmentation of semantic information within high-level features, concomitant with the enhancement of detailed information within low-level features; and 2) extraction of mid-level features tailored to fortify specific facets of objects. To realize this overarching aim, we present the GNM to seamlessly combine attributes from adjacent stages, as delineated in Fig. 3.

Notably, the GNM fuses the features extracted from different stages of the backbone using a residual learning scheme. To illustrate this with the features extracted from the t_1 image as an example: we apply max pooling and a 3×3 ConvLayer to the feature denoted as $f_2^{t_1}$, resulting in a down-sampled resolution denoted as $f_3^{t_1}$. This outcome is then processed through BatchN and ReLU activation functions, increasing its

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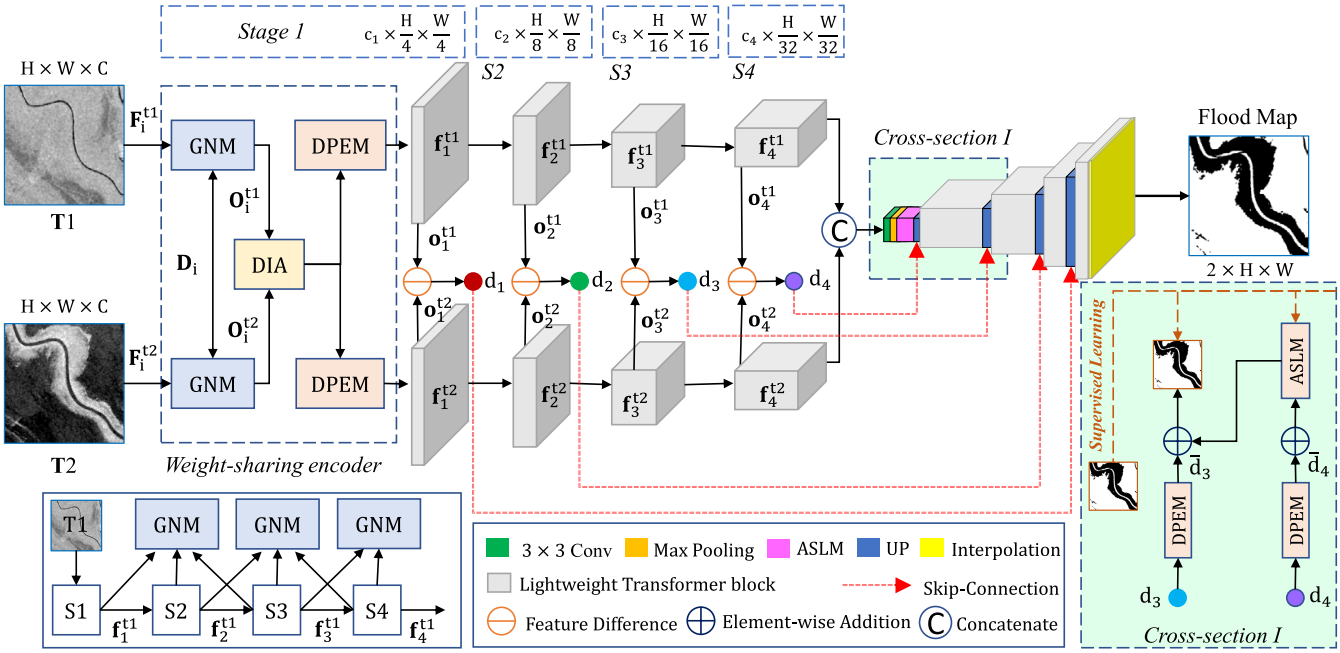


Fig. 2. Depicts the design of the proposed lightweight-aware transformer network (LiST-Net). Shared weights in the encoder's Siamese branches facilitate extracting temporal features by applying distinct convolutional operations. The GNM integrates temporal features from adjacent stages, enhancing their representation. To reduce the computational cost, a lightweight DIA module focuses on channel interaction. The DPEM optimizes distinctive features at each level, which are then merged into the decoder. Features from the ASLM are combined with the DPEM to polish changing water boundaries and produce a final change flood map.

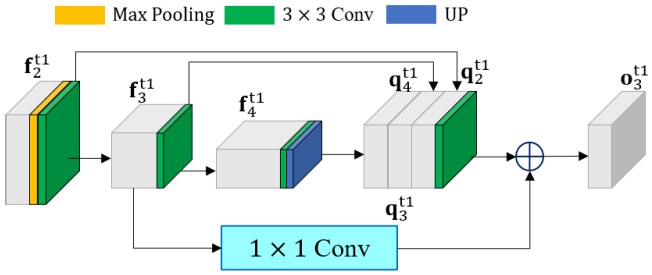


Fig. 3. Representation of the proposed GNM concept.

channels to obtain the derivation $q_2^{t_1}$. Simultaneously, $f_3^{t_1}$ is adjusted through a 3 × 3 ConvLayer to produce $f_4^{t_1}$, which is further refined using BatchN and ReLU to derive $q_3^{t_1}$. Similarly, $f_4^{t_1}$ is convolved via a 3 × 3 ConvLayer to reduce its channels, followed by bilinear upsampling to match the resolution of $q_4^{t_1}$. This comprehensive sequence of operations is succinctly formulated as follows:

$$\begin{aligned} q_2^{t_1} &= \mathrm{Conv}_{3\times3}\big(\mathrm{MaxP}\big(f_2^{t_1}\big)\big) \\ q_3^{t_1} &= \mathrm{Conv}_{3\times3}\big(f_3^{t_1}\big) \\ q_4^{t_1} &= \mathrm{UP}\big(\mathrm{Conv}_{3\times3}\big(f_4^{t_1}\big)\big). \end{aligned} \tag{5}$$

The subsequent phase involves the integration of features from adjacent stages, accomplished through the concatenation process, thereby aggregating multilevel semantic and detailed information. These merged features are then passed through a 3 × 3 ConvLayer to fine-tune the concatenated features. Employing the tenets of residual learning, a 1 × 1 ConvLayer is applied to adjust the channel dimensions of $f_3^{t_1}$, followed by the summation of these adjusted features through element-wise addition with the sequential features. This operation results in the generation of enhanced temporal features denoted as $o_3^{t_1}$. Analogously, this iterative process is replicated for features extracted from image $T2$, yielding the derivation of $o_3^{t_2}$. This entire operational sequence is succinctly expressed through the following formulation:

$$o_3^{t_1} = \mathrm{Conv}_{1\times1}\big(f_3^{t_1}\big) + \mathrm{Conv}_{3\times3}\big(\mathbf{Cat}\big(q_2^{t_1}; q_3^{t_1}; q_4^{t_1}\big)\big) \tag{6}$$

where $\mathbf{Cat}(\cdot)$ denotes a concatenation process.

C. DIA Module

With an escalation in spatial resolution for input images with dimensions HW, the computational complexity experiences quadratic growth, denoted as $\mathcal{O}((HW)^2)$. This escalation in complexity adds to the challenges of network training. In this study, we introduce the DIA module as a substitute for self-attention within the conventional transformer architecture, illustrated in Fig. 4. DIA relies on three fundamental concepts: Key ($\mathbb{K}$), Query ($\mathbb{Q}$), and Value ($\mathbb{V}$). These concepts are derived through convolutional projection across the depth of the input attribute. Specifically, we encode the feature maps O_i through channel–channel attention, generating an attention map $\mathbb{A}_1$. This spatial attention map exhibits linear complexity with a size of $\mathbb{R}^{(C\times C)}$, in contrast to the quadratic complexity of the traditional transformer's spatial attention map with a size of $\mathbb{R}^{(WH\times WH)}$. The attention map $\mathbb{A}_1$ is obtained using the dot product $\mathbb{Q}\mathbb{K}^T$ where $\mathbb{K}^T \in \mathbb{R}^{(WH\times C)}$ and $\mathbb{Q} \in \mathbb{R}^{(C\times WH)}$. Simultaneously, utilizing convolution, channel-width attention, and channel-height attention are encoded to produce attention

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maps $\mathbb{A}_2$ with size $\mathbb{R}^{(C\times W)}$ and $\mathbb{A}_3$ with size $\mathbb{R}^{(C\times H)}$, respectively. $\mathbb{A}_2$ and $\mathbb{A}_3$ enhance the interactions between the channel and its spatial dimensions. Subsequently, the three attention maps are combined, and the computational complexity of DIA becomes $\mathcal{O}(CC+CW+CH)$. This complexity is significantly lower than the traditional quadratic complexity of $\mathcal{O}((HW)^2)$. Formulations for attention maps are presented in (7), and experiments validating the effectiveness of the proposed DIA module are detailed in Section III

$$\begin{aligned}\mathbb{A}_1(\mathbb{Q},\mathbb{K},\mathbb{V}) &= \mathbb{S}\,\text{oftmax}\left(\frac{\mathbb{Q}\mathbb{K}^T}{\beta}\right)\mathbb{V}+O_i\\ \mathbb{A}_2(C,W) &= \mathbb{S}\,\text{igmoid}(\text{Conv}_{1\times1}(\text{Agg}(O_{r1})))O_{r1}\\ \mathbb{A}_3(C,H) &= \mathbb{S}\,\text{igmoid}(\text{Conv}_{1\times1}(\text{Agg}(O_{r2})))O_{r2}\\ \mathbb{A} &= \gamma_1\mathbb{A}_1+\gamma_2\mathbb{A}_2+\gamma_3\mathbb{A}_3 \end{aligned} \tag{7}$$

where $\mathbb{A}(\cdot)$ is the attention map, the parameter β, which is learnable, serves as a scaling factor governing the size of the dot product, O_i is the input feature, $\text{Conv}_{1\times1}$ denotes a 1×1 ConvLayer, $\text{Agg}(\cdot)$ signifies the aggregation process of average and max pool, the tensors $O_{r1}\in\mathbb{R}^{H\times C\times W}$ and $O_{r2}\in\mathbb{R}^{W\times C\times H}$ represent the reshaped dimensions of the input feature O_i, and $\gamma_1,\gamma_2,$ and γ_3 are the predefined hyper-parameters.

D. Detail-Preserving Enhancement Module

In contemporary CD methodologies, there is a tendency to overlook the significance of fine-grained features, ultimately leading to compromised integrity along the edges of changing water boundaries. One approach to address this issue involves utilizing a different image derived from multidate scenes to enhance the integrity of changed water edges. However, conventional algebraic operations like subtraction or the application of attention to augment features from multidate images incur substantial computational costs and exhibit limited capacity to represent features, thereby imposing constraints on optimization. This study introduces a novel lightweight DPEM aimed at comprehensively capturing different information from multidate images, as illustrated in Fig. 5. The DPEM comprises two distinct branches dedicated to learning fine-grained features: the detail-pooling branch and the difference enhancement branch. The difference-enhancement branch employs the lightweight spatio-channel 3-D attention module (LS3DAM) to generate more comprehensive change features. This module, specifically designed for SAR image-based flood mapping, combines 3-D spatial and channel attention to learn attention weights efficiently, inspired by [57]. Concurrently, the other branch focuses on detail enhancement through convolution and concatenation, applying LS3DAM to extract finer details, which is paramount for precise flood boundary delineation. Subsequently, the detail-pooling branch aggregates this detailed information through LS3DAM attention. Finally, as elucidated in Section III, both branches are seamlessly combined within the DPEM module, resulting in a highly effective double boost. This entire process is formulated as follows:

$$f^e_{p1} = \text{Conv}_{1\times1}\left(\mathbf{Cat}\left(\text{Conv}_{3\times3}\left(f_i^{t_1}\right);\text{Conv}_{3\times3}\left(f_i^{t_2}\right)\right)\right) \tag{8}$$

$$f^e_{p_2} = \text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(f_i^{t_1}\right)\oplus\text{Conv}_{3\times3}\left(f_i^{t_2}\right)\right) \tag{9}$$

$$\begin{aligned} f^e_{\text{aggr}} &= \boldsymbol{A}_s\left|\boldsymbol{A}_s\left(f^e_{p_1}\right)\oplus\boldsymbol{A}_s\left(f^e_{p_2}\right)\right|\\ f^e_{\text{diff}} &= \boldsymbol{A}_s\left|\boldsymbol{A}_s\left(\text{Conv}_{3\times3}\left(f_i^{t_1}\right)\right)\ominus\boldsymbol{A}_s\left(\text{Conv}_{3\times3}\left(f_i^{t_2}\right)\right)\right|\\ f^e_{\text{DPEM}} &= f^e_{\text{aggr}}\oplus f^e_{\text{diff}} \end{aligned} \tag{10}$$

where $f^e_{p_1}$ and $f^e_{p_2}$ denote the enhanced features of the two paths in the detail-pooling branch, $\text{Conv}_{1\times1}$ and $\text{Conv}_{3\times3}$ signify a 1×1 and 3×3 convolution, respectively, $f_i^{t_1}$ and $f_i^{t_2}$ represent the multidate features at the ith stage, **Cat** denotes the concatenation process, f^e_{aggr} and f^e_{diff} are the enhanced features of the detail-pooling and difference-enhancement branches, respectively, $\oplus$ and $\ominus$ are the element-wise addition and subtraction, respectively, the operation $|\cdot|$ signifies the absolute value, and $\boldsymbol{A}_s$ refers to the lightweight simple attention.

E. Attentive Supervised Learning Module

Higher level features often necessitate contextual guidance for processing noise during feature fusion. The ASLM introduced in this article fulfills this role. Fig. 6 illustrates the ASLM's contribution to enhancing feature fusion. To elucidate, we consider the DPEM output of the last stage, denoted as $\bar{d}_4$, as an example, representing the ASLM input. Initially, we generate the contextual information of the changing water map $\boldsymbol{M_4}$ by applying a 1×1 ConvLayer followed by a Sigmoid function to the input feature. Subsequently, an Invert LUT is applied to the original changing map $\boldsymbol{M_4}$ to produce the reversed change map $\boldsymbol{M_4^r}$ (i.e., $\boldsymbol{M_4^r}=\text{InvertLUT}(\boldsymbol{M_4})$), which encompasses the unchanged background parameters. Following this, a 1×1 ConvLayer is applied to both $\boldsymbol{M_4}$ and $\boldsymbol{M_4^r}$, resulting in a pixel-wise mask $\boldsymbol{G_4}$. This mask serves as a gateway for changing water information, guiding the learning process of $\bar{d}_4$. Through the element-wise multiplication of gate $\boldsymbol{G_4}$ with feature $\bar{d}_4$ and subsequent application of a 3×3 ConvLayer, we obtain the attentively supervised learning feature $\bar{d}_4^r$. This procedure is repeated on features from other levels and combined with the DPEM outputs, resulting in efficient feature fusion for CD. The proposed approach is formulated as follows:

$$\begin{aligned} \bar{d}_4 &= f^{e4}_{\text{DPEM}}\\ \boldsymbol{M_4} &= \text{Sigmoid}\left(\text{Conv}_{1\times1}(\bar{d}_4)\right)\\ \boldsymbol{G_4} &= \text{Conv}_{1\times1}\left(\mathbf{Cat}\left(\boldsymbol{M_4};\boldsymbol{M_4^r}\right)\right)\\ \bar{d}_4^r &= \text{Conv}_{3\times3}(\bar{d}_4\otimes\boldsymbol{G_4}). \end{aligned} \tag{11}$$

Here, $\text{Conv}_{1\times1}$ and $\text{Conv}_{3\times3}$ denote 1×1 and 3×3 convolutions, respectively. **Cat** signifies the concatenation process, and $\otimes$ denotes element-wise multiplication.

F. Mixed Loss Function

In the context of the CD task, there is often a significantly higher ratio of unchanged regions compared to changed ones, leading to a class imbalance challenge. To address this challenge and enhance the network's ability to learn from intricate

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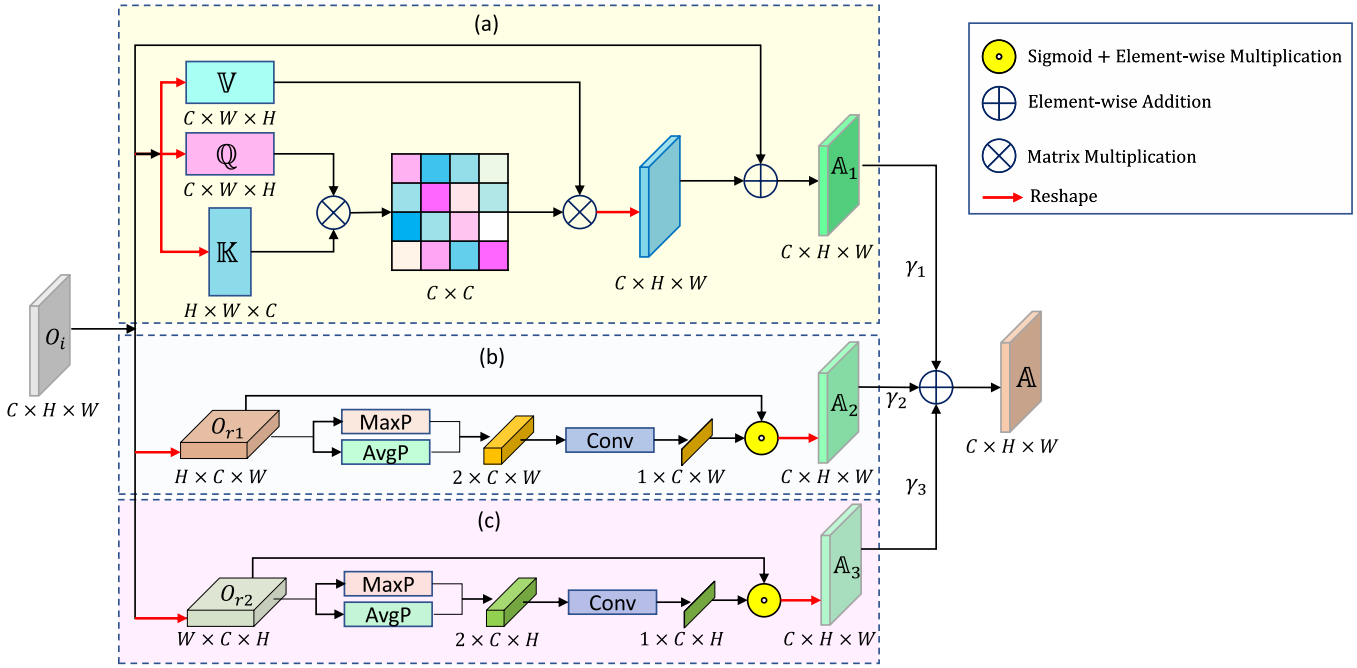


Fig. 4. Illustration of the proposed DIA module. (a) Depicts the channel–channel attention branch to obtain $\mathbb{A}_1$, (b) illustrates the channel-width attention branch to obtain $\mathbb{A}_2$, and (c) represents the channel-height attention branch to obtain $\mathbb{A}_3$.

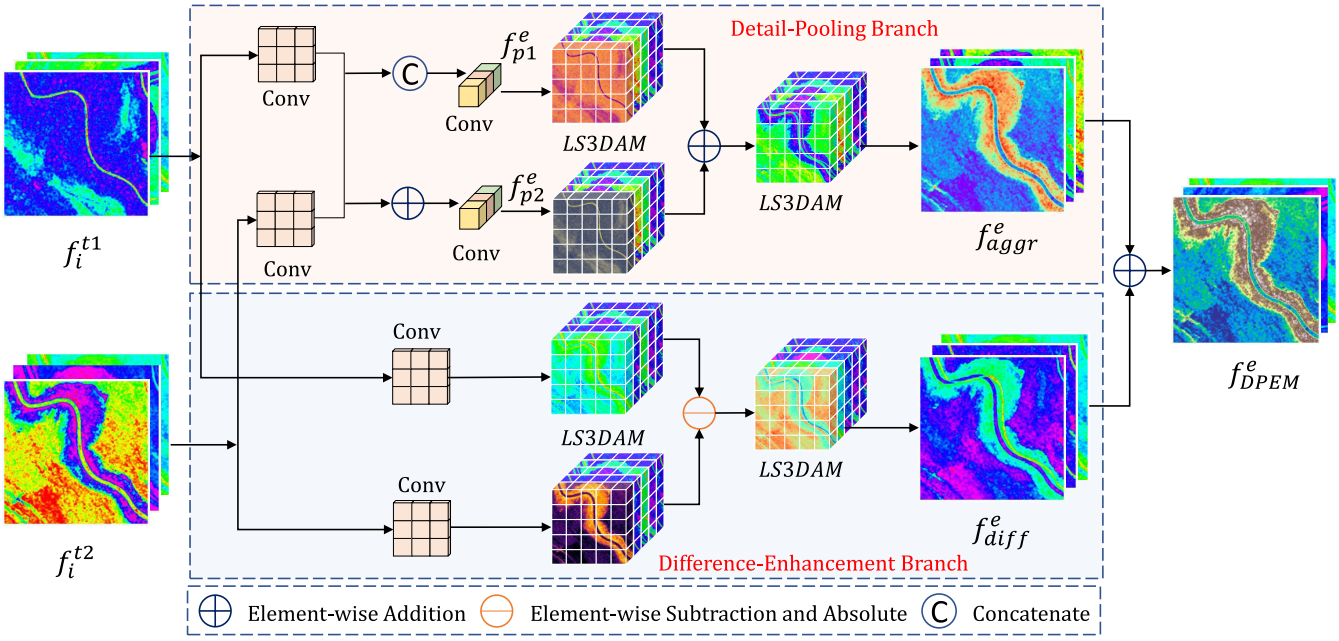


Fig. 5. Depicts an illustration of the proposed DPEM module.

scenes, we employ a combination of binary loss, denoted as $\mathcal{L}_{\text{bce}}$, along with dice loss, denoted as $\mathcal{L}_{\text{dice}}$, as depicted in (14)

$$\mathcal{L}_{\text{bce}}(\text{p}, \text{g}) = (1 - \text{g}) \odot \log(1 - \text{p}) + \text{g} \odot \log \text{p} \tag{12}$$

$$\mathcal{L}_{\text{dice}}(\text{p}, \text{g}) = 1 - \frac{2 \odot \text{p} \odot \text{g}}{|\text{p}| + |\text{g}|} \tag{13}$$

$$\mathcal{L}(\text{p}, \text{g}) = \mathcal{L}_{\text{bce}}(\text{p}, \text{g}) + \mathcal{L}_{\text{dice}}(\text{p}, \text{g}) \tag{14}$$

where g is the truth data, p signifies the predicted change map, $\odot$ represents a dot-product process, and $|\cdot|$ denotes the ℓ_1 norm.

IV. Experiments

A. Dataset and Assessment Criteria

In our experiments, we conducted experiments on two extensive publicly available flood detection SAR datasets. The detailed information is listed as follows.

1) *S1GFloods [15]:* The S1GFloods dataset comprises a total of 5360 pairs of Sentinel-1 SAR images, characterized by a resolution of 10 m and a patch size of 256 × 256. These images cover 46 locations worldwide. To ensure a balanced distribution, we randomly allocated

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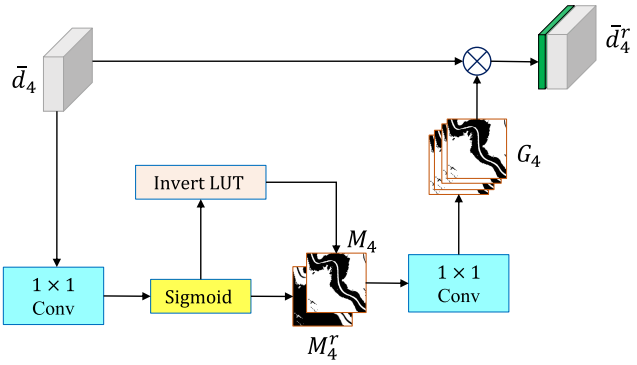


Fig. 6. Illustrates the proposed ASLM.

4.3k, 0.53k, and 0.53k patches for training, validation, and testing purposes, respectively. The ratios of training, validation, and testing stand at 6:2:2.

2) *ETCI-2021 [16]:* The ETCI-2021 dataset encompasses Sentinel-1 SAR images that span across five distinct geographic regions. It consists of a remarkable collection of over 68 000 patches, each sized 256 × 256. Within this dataset, we utilized 33k, 10.4k, and 24.6k pairs for training, validation, and testing, respectively. To mitigate the risk of over-fitting in this study, data enhancement processes such as random flip, histogram matching, random rotation, and clipping techniques were applied.

3) *Evaluation Metrics:* To comprehensively assess the network's performance on the flood benchmark datasets, six principal evaluation metrics were employed [11]: 1) precision ($\mathcal{P}$), which measures the rate of true positives against all positive predictions made by the model; 2) recall ($\mathcal{R}$), which evaluates the model's accuracy in correctly identifying relevant flood cases; 3) F-score ($F1$), which represents the harmonic mean of precision and recall, providing a balance between them; 4) intersection over union (IoU), which quantifies the overlap between the predicted flood map and the ground truth, reflecting the accuracy of the model's spatial delineation; 5) kappa coefficient ($\mathcal{K}$), which indicates the reliability of the detection map by gauging interrater agreement for categorical assessments; and 6) overall error (OE), which indicates the ratio between the number of incorrectly detected pixels in the result and the total number of the ground truth map. In addition, overall accuracy (OA) measures the proportion of correctly detected pixels relative to the total number of pixels in the ground truth map. The detailed definitions of the above evaluation metrics are formulated as follows:

$$\begin{aligned}
\text{OA} &= \frac{\mathcal{TP}+\mathcal{TN}}{\mathcal{TP}+\mathcal{FN}+\mathcal{FP}+\mathcal{TN}} \\
\text{OE} &= \frac{\mathcal{FP}+\mathcal{FN}}{\mathcal{TP}+\mathcal{FN}+\mathcal{FP}+\mathcal{TN}} \\
\text{IoU} &= \frac{\mathcal{TP}}{\mathcal{TP}+\mathcal{FP}+\mathcal{FN}} \\
\mathcal{R} &= \frac{\mathcal{TP}}{\mathcal{TP}+\mathcal{FN}} \\
\mathcal{P} &= \frac{\mathcal{TP}}{\mathcal{TP}+\mathcal{FP}} \\
F1 &= \frac{2}{\text{P}^{-1}+\text{R}^{-1}} \\
\mathcal{K} &= \frac{\text{OA}-\text{PE}}{1-\text{PE}} \\
\text{PE} &= \frac{\mathcal{TN}^2+2\mathcal{TN}(\mathcal{FP}+\mathcal{FN})+\mathcal{FN}^2+\mathcal{TP}^2}{(\mathcal{TP}+\mathcal{FP}+\mathcal{TN}+\mathcal{FN})^2}
\end{aligned} \tag{15}$$

where $\mathcal{FN}$, $\mathcal{TN}$, $\mathcal{FP}$, and $\mathcal{TP}$ stand for false negatives, true negatives, false positives, and true positives, respectively.

B. Operational Information

The PyTorch toolbox [58] is employed in this study to implement the proposed method, with training/validation extending over 100 epochs on NVIDIA's RTX 3090 (32 GB GPU). Throughout the training iteration, the backbone of the proposed method is initialized with ViTAEv2 [56] parameters pretrained on MillionAID [59] dataset for preliminary feature extraction. We employed an AdamW optimizer with a weight-decay set to 0.0006, a starting rate of learning of 0.0001, a momentum of 0.99, and a batch size of 8. The optimal experimental values for γ_1:γ_2:γ_3 were determined to be 2:1:1. To address the challenge posed by the class imbalance problem, this article mixes binary loss and dice loss to optimize the model.

C. Comparison With Leading Techniques

Here, we have chosen eight cutting-edge CD approaches for comparison with our proposed LiST-Net method. These include five CNN-based approaches: FC-Diff and FC-Cat [35], STA-Net [7], AFDE-Net [8], and SNU-Net [9], and three vision transformers-based approaches: SwinSU-Net [14], BIT [12], and ChangeFormer [13]. To ensure fair comparisons, we have reimplemented these approaches using modified codes based on their proposed parameters. In addition, we provide the computational expenses and model parameters of these approaches for reference.

1) Quantitative Evaluation: Table I briefly compares our LiST-Net model against eight advanced methods using key performance metrics on two flood detection datasets. The proposed LiST-Net model demonstrates notable improvements across all metrics, particularly with a 7.3% increase in $F1$, a 12.2% increase in IoU, and a 9.9% increase in the $\mathcal{K}$ over the ETCI-2021 dataset. LiST-Net achieved the lowest OE of 2.4%, reflecting its precision in identifying change regions as shown in Fig. 7. On the S1GFloods dataset, LiST-Net showcases its generalizability, achieving a significant 7.4% enhancement in IoU and an OE of 2.2%, which indicates a substantial reduction in the average false and missed alarms. These results affirm that the proposed LiST-Net model can accurately capture changing spatial features without overlooking critical details. In addition, LiST-Net contained relatively the lowest number of parameters (Pms) of 1.23× (7.34M) and has a low computational cost (FLPs) of 9.31× (11.78G) with a relatively higher $F1$ of 94.7% among the compared

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TABLE I

QUANTITATIVE COMPARISONS ON TWO FLOOD DETECTION DATASETS IN $\mathcal{P}$, $\mathcal{R}$, $F1$, IOU, $\mathcal{K}$, AND OE TERMS. THE BEST OUTCOME IS HIGHLIGHTED IN RED, THE SECOND IN BLUE, AND THE WORST IN GREEN. THE SYMBOL ↓ INDICATES AN INCREASE IN IMPROVEMENT

Scenario	Method	ETCI-2021 [16]						S1GFloods [15]					
		$\mathcal{P}$ %	$\mathcal{R}$ %	F1 %	IoU %	$\mathcal{K}$ %	OE %↓	$\mathcal{P}$ %	$\mathcal{R}$ %	F1 %	IoU %	$\mathcal{K}$ %	OE %↓
CNN	FC-Diff [35]	87.2	67.0	75.8	63.7	73.3	6.7	86.6	94.1	90.2	81.3	89.3	4.7
	FC-Cat [35]	87.9	67.1	76.1	63.5	73.8	6.2	89.4	94.1	91.7	81.8	90.1	4.0
	STA-Net [7]	88.8	73.3	80.3	65.2	77.2	5.4	91.5	93.3	92.4	83.7	91.8	3.8
	AFDE-Net [8]	89.4	73.9	80.9	65.8	76.4	5.2	93.8	94.8	94.3	85.4	93.2	2.9
	SNU-Net [9]	89.7	76.4	82.5	70.1	78.7	3.0	94.6	95.2	94.9	87.2	93.5	2.3
ViT	SwinSU-Net [14]	86.9	77.1	81.7	72.4	80.9	3.5	94.0	92.8	93.4	86.5	91.7	3.8
	BIT [12]	88.5	76.7	82.2	73.7	82.4	3.0	93.9	93.5	93.7	86.2	92.4	3.1
	ChangeFormer [13]	89.3	76.5	82.4	74.1	82.7	2.7	94.3	93.9	94.1	87.3	93.5	2.5
	LiST-Net (Ours)	89.5	77.6	83.1	75.9	83.2	2.4	95.6	93.8	94.7	88.7	93.8	2.2

TABLE II

RUNNING EFFICIENCY AND PARAMETERS WITH $F1$-SCORE ON THE S1GFLOODS DATASETS

Method	FLPs (G)	Pms. (M)	Time (m:s)	F1 (%)
FC-Diff [35]	9.450	1.350	02:58	90.2
FC-Cat [35]	10.62	1.550	02:52	91.7
STA-Net [7]	26.03	16.92	06:15	92.4
AFDE-Net [8]	75.15	16.33	25:40	94.3
SNU-Net [9]	109.69	09.03	14:35	94.9
SwinSU-Net [14]	31.30	28.20	22:49	93.4
BIT [12]	25.20	24.40	11:22	93.7
ChangeFormer [13]	16.84	24.30	08:20	94.1
LiST-Net (Ours)	11.78	7.340	10:30	94.7

methods as shown in Table II and Fig. 8. Consequently, a pressing requirement exists for crafting a lightweight strategy with lower computational costs, decreased parameters, and the ability to effectively handle more challenging scenarios while maintaining excellent performance, as demonstrated in our proposed LiST-Net method.

2) Qualitative Evaluation: Visual comparisons, as shown in Figs. 9 and 10, illustrate the superiority of the proposed LiST-Net over most state-of-the-art methods that have limited boundary recognition ability. For instance, LiST-Net more accurately identifies flood-affected areas, as evidenced by the precise delineation of the flooded island within the Nile River [see Fig. 9(a)]. In contrast, other methods often yield misclassifications, as observed in Figs. 9(b) and 10(c), where they fail to distinguish between actual flooding and similar-looking features. Although the BIT and ChangeFormer models show relatively similar results to LiST-Net, they are quantitatively outperformed across various flood event data and scenarios. Consequently, we assert that LiST-Net effectively reduces false positives and aligns closely with the ground truth, which is essential for reliable flood mapping.

D. Efficiency Analysis

Fig. 11(a) and (b) presents quantitative comparisons of various models on the two flood detection datasets. The performance metrics of our model surpass those of the other approaches. Precisely, on the ETCI2021 datasets, the LiST-Net exhibits a 12.2% increase in IoU and a 7.3% improvement in $F1$ score compared to the baseline FC-Diff model. Furthermore, our model achieves the highest performance, with a 4.5% increase in $F1$ and 7.4% in IoU within S1GFloods datasets. In addition, Fig. 11(c) and (d) illustrates the distribution of accuracy indices using a boxplot. Across both datasets, the boxplots indicate that all accuracy indices of our model are significantly higher than those of other models, which exhibit numerous outliers. In Fig. 12, the loss performance during the training of various networks is depicted. Notably, the training loss associated with our proposed LiST-Net method exhibits a discernible trend, gravitating toward convergence within the initial 20 epochs. In contrast, SwinSU-Net displays a slower convergence rate, even with a gradual extension of the training duration, attributed to the utilization of contrastive loss instead of binary cross-entropy. Moreover, Fig. 13 and Table II present the training time in minutes for each epoch and model parameters, aiming to assess the efficiency of our LiST-Net on the S1GFloods datasets. The LiST-Net demonstrated approximately 4.0% less training time than the SNU-Net method, despite their proximity in terms of parameter count. Notably, more intricate methods like AFDE-Net and SwinSU-Net exhibited considerably longer training times per epoch, attributable to their significantly higher number of parameters. Furthermore, our method displayed a 2.0% increase in training time compared to the ChangeFormer method, yet it achieved a 1.0% higher $F1$ score, highlighting the commendable performance of the proposed approach.

E. Ablation Study

We executed ablation studies to assess the impact of the proposed GNM, DIA, DPEM, and ASLM components, with the corresponding quantitative statistics presented in Table III. Evaluating the impact of the GNM component on our model involved its removal from the baseline, resulting in decreased evaluation metrics in both datasets. Specifically, the IoU experienced a decline of approximately 2.2% and 2.9% in ETCI-2021 and S1GFloods, respectively. This emphasizes the crucial role of aggregating features from adjacent stages within the backbone in enhancing the representativeness of the extracted features. Furthermore, the results reveal that introducing the DIA component to the baseline significantly improves detection accuracy, with increases of about 6.4%

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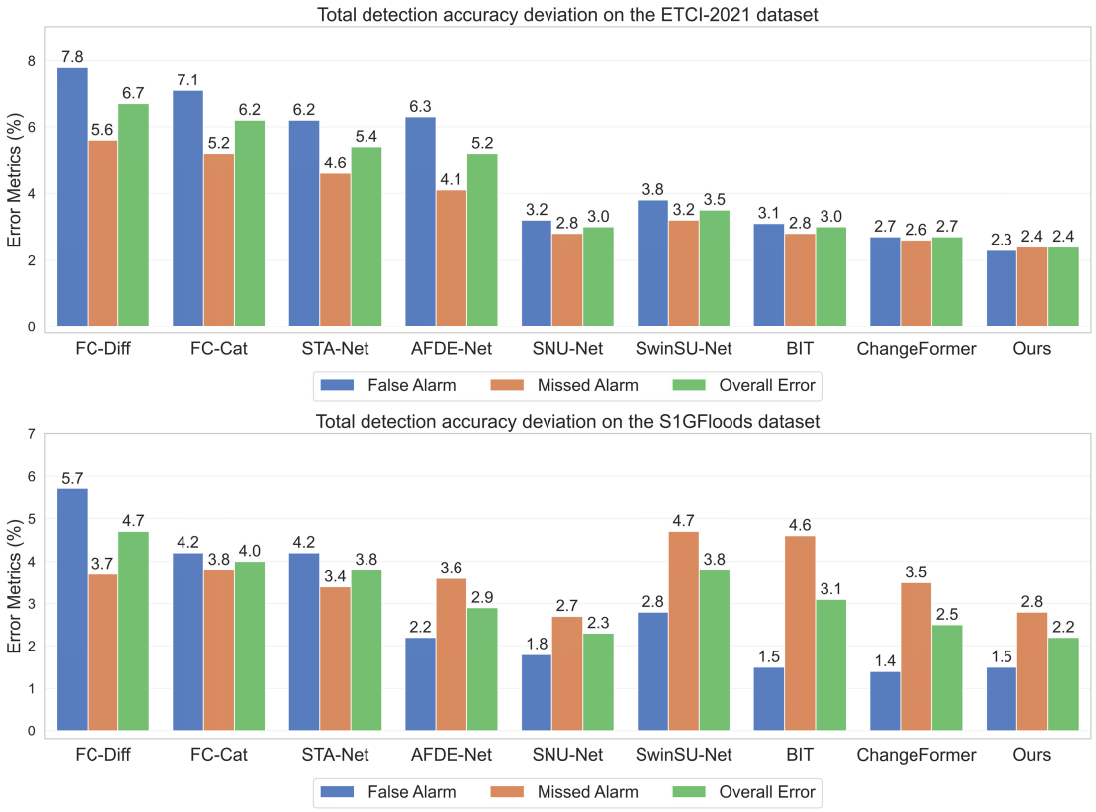


Fig. 7. Total detection accuracy deviation on the ETCI-2021 and S1GFloods dataset with error metrics.

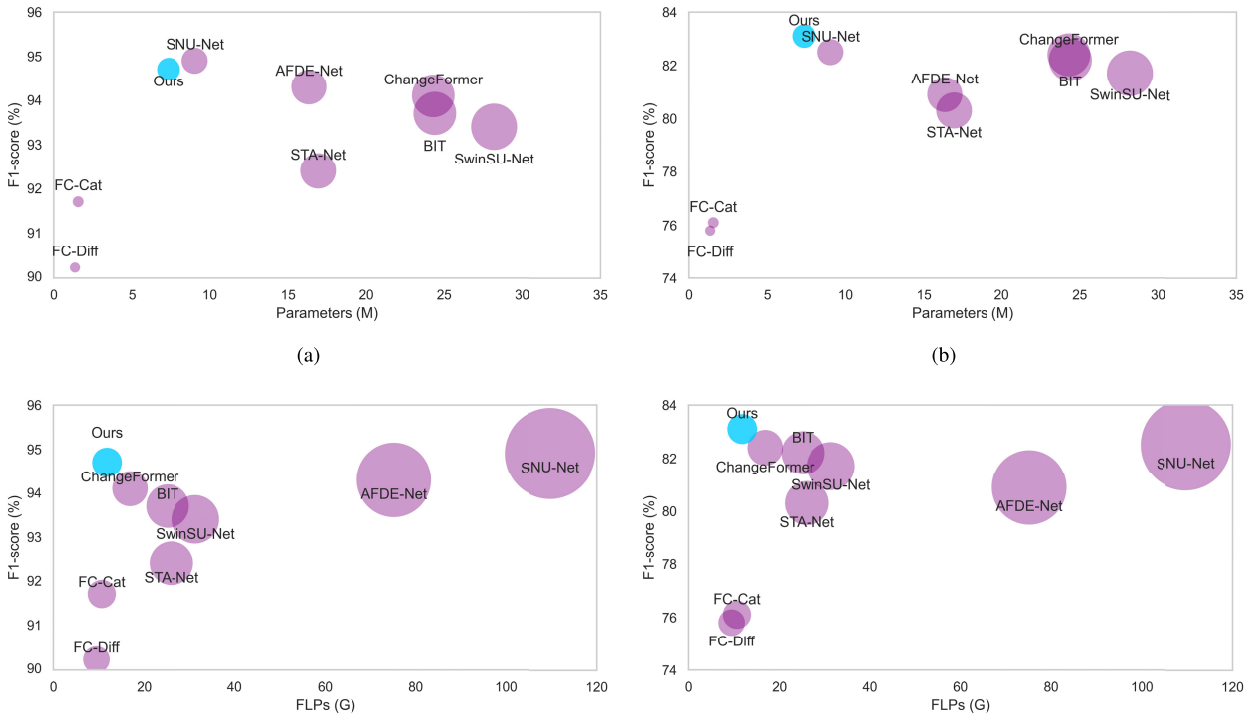


Fig. 8. Comparison of model complexity among various methods concerning parameters (Pms), computational cost (FLPs), and $F1$ value on both datasets. (a) and (c) S1GFloods dataset, while (b) and (d) correspond to the ETCI-2021 dataset.

and 6.9% concerning IoU on the ETCI-2021 and S1GFloods datasets, respectively. Notably, the DIA component not only enhances detection accuracy but also reduces computational complexity by almost half, thanks to its ability to combine interactive attention mechanisms at the dimensional level for more effective information capture. In addition, the inclusion

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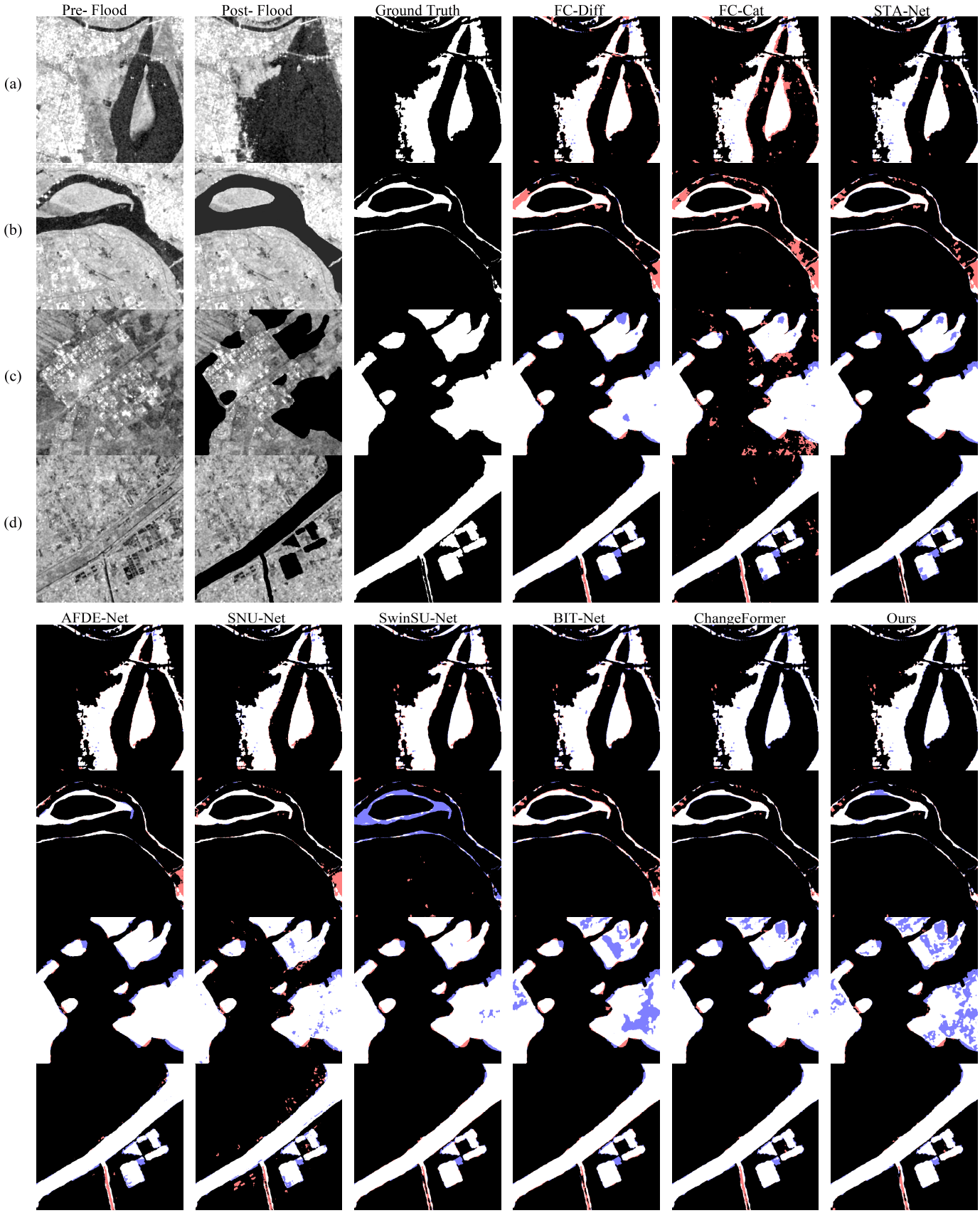


Fig. 9. Qualitative comparison within S1GFloods datasets. (a)–(d) Four different regions of the dual-date SAR imagery selected from the S1GFloods test set to compare our LiST-Net method with other excellent methods. Rendered colors: white and red for true and false positives, black and blue for true and false negatives, respectively.

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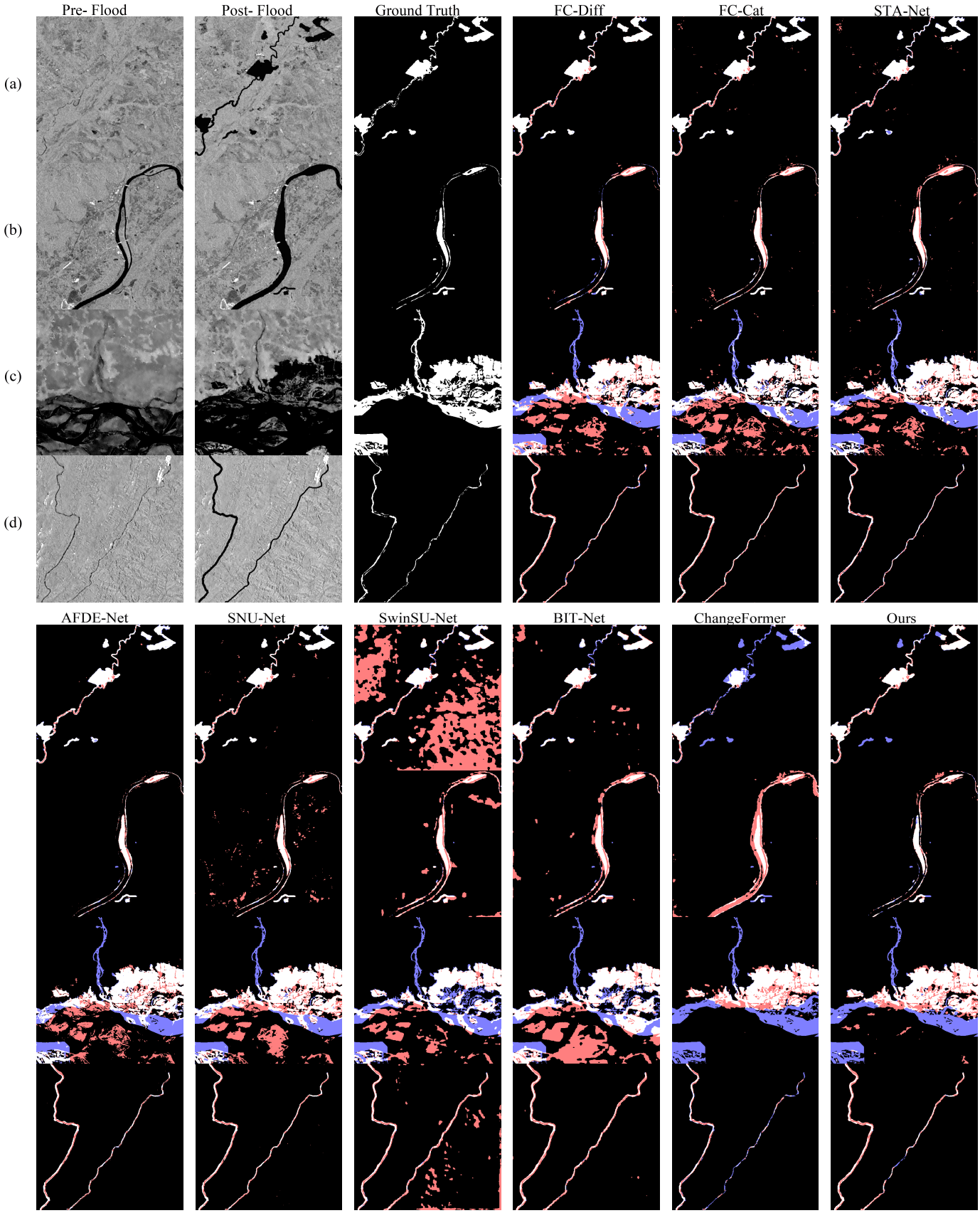


Fig. 10. Qualitative comparison within ETCI2021 datasets. (a)–(d) Four different regions of the dual-date SAR imagery selected from the ETCI2021 test set to compare our LiST-Net method with other excellent methods. Rendered colors: white and red for true and false positives, black and blue for true and false negatives, respectively.

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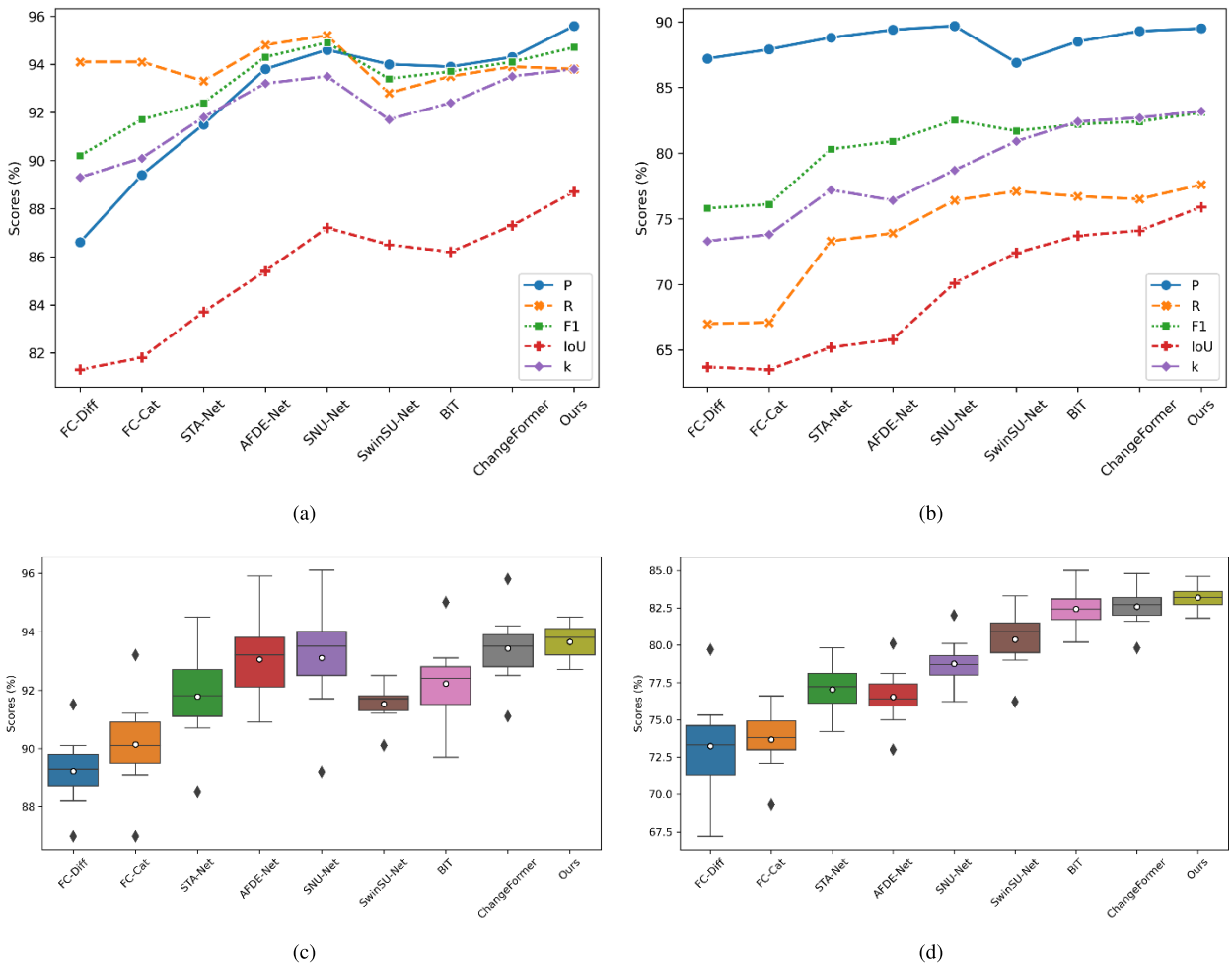


Fig. 11. Depicts quantitative comparisons and boxplots for accuracy indices on the two datasets. (a) and (c) Display accuracy indices for the S1GFloods dataset, whereas (b) and (d) present those for the ETCI2021 dataset.

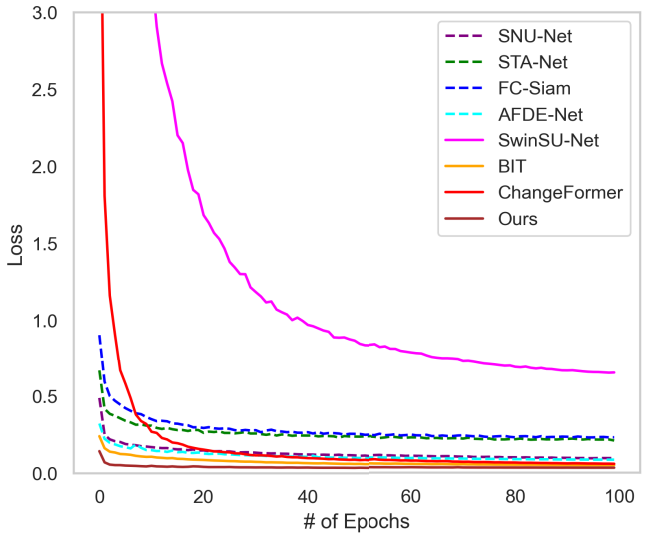


Fig. 12. Performance concerning loss and convergence during the training of the proposed network.

of the DPEM component contributes to a modest improvement in detection accuracy. ASLM, by aggregating features from coarse to fine, leads to a slight performance increase. Moreover, the combined usage of GNM, DPEM, and ASLM enhances $F1$ performance by 3.4% and 0.7% compared to the baseline with the ETCI-2021 and S1GFloods datasets, respectively. Notably, the addition of the DIA component to this combination results in a significant performance improvement of 7.6% and 7.1% in IoU with ETCI-2021 and S1GFloods data, respectively. This improvement is attributed to a more comprehensive capture of change information and effective integration of deep semantic features.

Table IV exhibits the experimental findings of hyperparameters γ_1, γ_2, and γ_3 within the DIA module across the two datasets. The findings reveal consistent performance across minor variations in these parameters, indicating their suitability for practical applications. We explored six intervals between 0.2 and 2.0 for γ_1, while γ_2 and γ_3 were investigated within intervals from 0.1 to 1.0, to ensure that the values of 2:1:1 are the optimal choice. The results were thoroughly analyzed using four indicators. Notably, it is observed that the values of ($\mathcal{R}$) gradually decrease with increasing hyperparameters, whereas ($\mathcal{P}$) shows an increasing trend. This suggests that predictions derived from higher parameters exhibit greater accuracy but cover fewer instances of ground truth. We deduce that these values prove effective for our application, striking a balance between the model's recall and accuracy.

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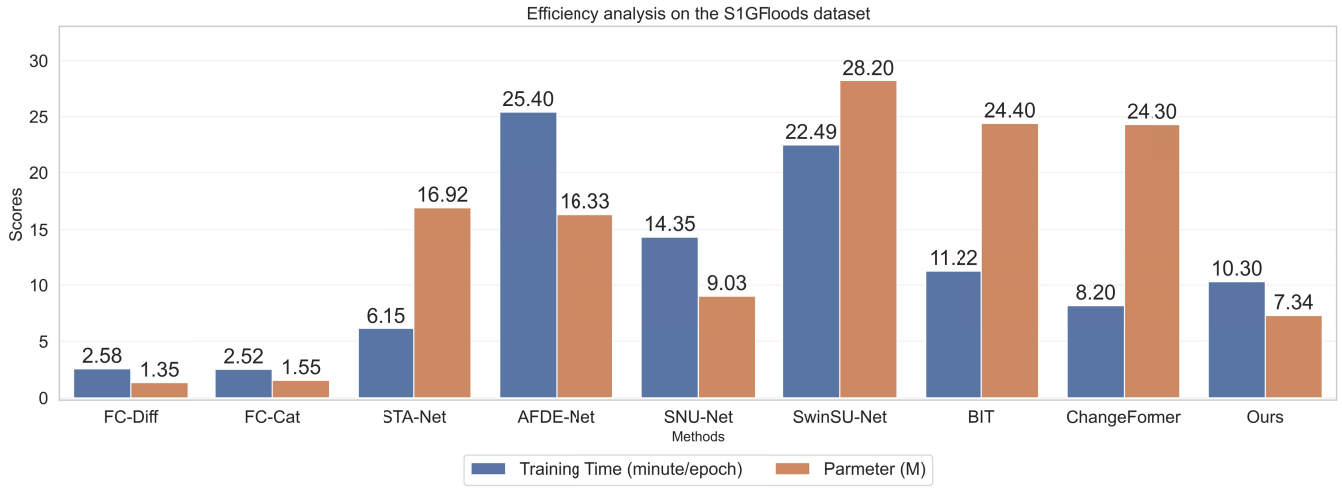


Fig. 13. Demonstration of an efficiency analysis involving cutting-edge techniques.

TABLE III

ABLATION EXPERIMENTS OF OUR PROPOSED MODULES ON THE TWO FLOOD DATASETS. THE ✓SYMBOL DENOTES THE INCLUSION OF THIS COMPONENT INTO THE PROPOSED LIST-NET METHOD. THE SHADE OF RED SERVES AS AN INDICATOR OF THE TOP-RANKING RESULT, WHILE THE UNDERLINING IS EMPLOYED TO DENOTE THE SECOND-HIGHEST RESULT

Method	GNM	DIA	DPEM	ASLM	ETCI2021				S1GFloods				FLPs	Pms
					$\mathcal{P}$ (%)	$\mathcal{R}$ (%)	F1 (%)	IoU (%)	$\mathcal{P}$ (%)	$\mathcal{R}$ (%)	F1 (%)	IoU (%)	(G)	(M)
Baseline +					84.1	69.7	76.2	68.3	90.3	90.5	90.4	81.6	17.84	5.83
	✓				86.9	72.1	78.8	70.5	91.7	93.3	92.5	84.5	17.73	5.83
		✓			88.3	78.1	82.9	74.7	95.7	93.9	94.8	88.5	8.83	5.17
			✓		87.2	73.7	79.9	71.8	93.7	88.6	91.1	82.3	16.74	5.92
				✓	88.7	74.1	80.7	73.3	92.4	90.1	91.2	82.2	15.46	6.94
		✓	✓	✓	89.6	76.3	82.4	74.1	93.2	91.8	92.5	84.7	18.68	7.23
	✓		✓	✓	88.5	72.3	79.6	71.5	93.3	89	91.1	82.4	19.71	7.12
	✓	✓		✓	89.1	76.3	82.2	74.2	95.5	91.8	93.6	85.8	11.37	6.63
	✓	✓	✓		88.4	73.7	80.4	72.7	95.3	92.9	94.1	87.1	11.18	6.74
	✓	✓	✓	✓	89.5	77.6	83.1	75.9	95.6	93.8	94.7	88.7	11.78	7.34

TABLE IV

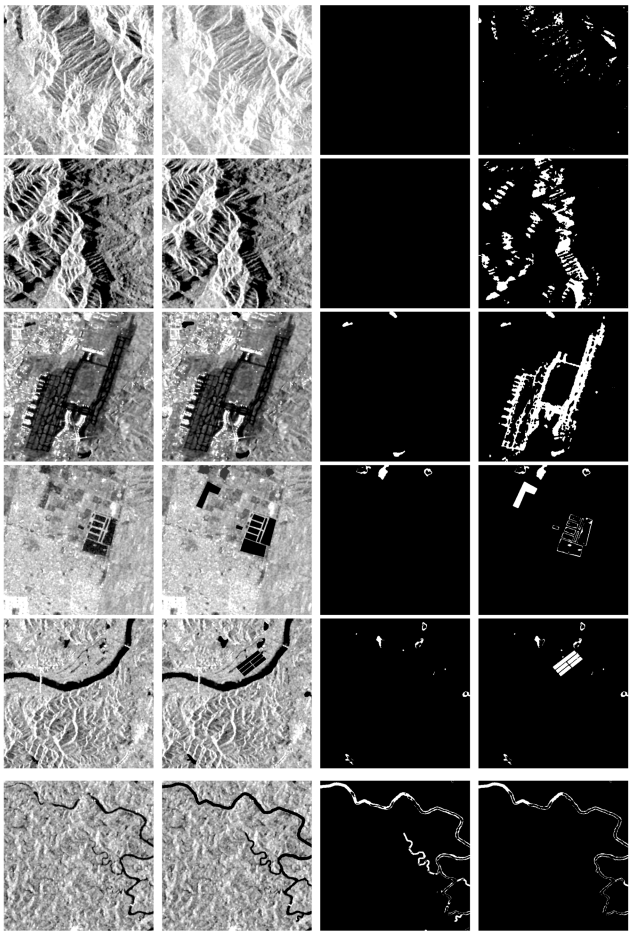
COMPARISON OF DETECTION PERFORMANCE WITH VARIOUS γ VALUES ON THE TWO BENCHMARK DATASETS. THE RED INDICATES THE OPTIMAL RESULT

Method	$\gamma 1$	$\gamma 2$	$\gamma 3$	ETCI-2021				S1GFloods			
				$\mathcal{P}$	$\mathcal{R}$	F1	IoU	$\mathcal{P}$	$\mathcal{R}$	F1	IoU
Base + DIA	0.2	0.1	0.1	85.2 %	78.8 %	81.9 %	72.3 %	87.3 %	98.8 %	92.7 %	83.1 %
	0.4	0.3	0.3	85.7 %	78.6 %	82.0 %	72.9 %	88.2 %	97.9 %	92.8 %	83.4 %
	0.6	0.4	0.4	86.5 %	78.2 %	82.1 %	73.2 %	89.5 %	96.6 %	92.9 %	83.5 %
	1.0	0.6	0.6	87.4 %	77.6 %	82.2 %	73.7 %	89.9 %	96.4 %	93.0 %	84.0 %
	1.5	0.8	0.8	88.1 %	77.2 %	82.3 %	74.3 %	90.1 %	96.3 %	93.1 %	84.7 %
	2.0	1.0	1.0	88.3 %	78.1 %	82.9 %	74.7 %	95.7 %	93.9 %	94.8 %	88.5 %

F. Limitations

Although our LiST-Net method attained exceptional performance on SAR datasets in detecting flooded areas in regular scenarios, there is still room for improvement in several aspects. It can be found that some cases have many false alarms and need to be improved. Fig. 14 (first-third row) shows a pair of images covered by towering mountains in China, making it easy to confuse bodies of water with shadows of hills that appear as dark, almost water-like areas. Furthermore, some areas alongside the layover also easily lead to false alarms due to visual properties similar to bodies of water. In addition, areas of airports that have low-intensity analogous to bodies of water in SAR scenes may give us suboptimal results and lead to false alarms. These challenges can be well addressed if it is possible to introduce a unique class token that minimizes the interference of unwanted changes. On the other hand, we find that many nonwater-related areas (such as agriculture) are detected as floodwaters through our proposed method. However, the datasets used in this article mainly focus on detecting flooded areas and ignore other categories when developing ground truth annotations. Intuitively, the proposed method can detect cropping boundaries and changing riverside boundaries due to rising flood levels with higher accuracy, as shown in Fig. 14 (fourth-sixth row). In addition, since there is a gap between optical and radar images such as texture and color, this will likely lead to a suboptimal result. Therefore, some pretrained models like [15] can be used to help capture high-level semantic information, enabling the model to concentrate solely on detecting changes within the

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(a) (b) (c) (d)

Fig. 14. Visual of some limited detection outcomes of our proposed model. (a) Time 1 scenes. (b) Time 2 scenes. (c) Binary ground truth. (d) Our proposed LiST-Net model.

region of interest. Therefore, we leave this point to our future work and further empirical exploration in this regard.

V. Conclusion

In this study, we propose a novel lightweight LiST-Net designed to enhance the accuracy of flood detection from SAR imagery. LiST-Net comprises the GNM, the DIA module, and the DPEM. The GNM leverages neighboring pixels to enhance detailed information and feature extraction ability, the DIA effectively reduces computational complexity, and the DPEM captures different information from multidate images at various feature levels. In addition, an ASLM is implemented to reduce noise through the pixel mask gate. The features derived from the DPEM and ASLM modules are combined to improve the accuracy of water edge delineation, yielding the final change flood map. To assess LiST-Net's performance, we conducted a series of quantitative, qualitative, and ablation studies. Experiments on extensive flood detection datasets, S1GFloods, and ETCI-2021, demonstrate that LiST-Net outperforms existing methods, achieving a 94.7% and 83.1% improvement in $F1$ on the S1GFloods and ETCI-2021 test sets, respectively. Furthermore, computational complexity and parameter analyses on LiST-Net's components reveal that DIA achieves lower computational costs (8.83G) and fewer parameters (5.17M), establishing it as a potent lightweight baseline in flood detection applications. Areas affected by layover and covered by shadows require further experimental exploration and improvement, which we leave as a focal point for future work.

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